

The Arf Switches Off: Quadratic Forms over the Full Nimber Field

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Abstract

Conway's full nimber field On_2 is algebraically closed of characteristic 2. Consequently the Artin–Schreier map is surjective, every regular finite-dimensional quadratic form over On_2 is hyperbolic, and the quadratic Witt group is zero. A possibly singular form is classified by the rank of its polar form, the radical dimension, and whether its restriction to the radical vanishes; there is no further Arf coordinate. The Clifford/Brauer–Wall class of every regular form is therefore split.

The finite-field Arf bit remains correct as a *relative* invariant over the chosen finite nim-subfield. It dies after a quadratic scalar extension, so its directed colimit through the finite nim-subfields is zero. This also exposes an empty mathematical middle case: every ordinal below ω^ω is algebraic over $\mathbb{F}_2$, so every finite metric in that segment has a common finite field of definition. The backend detects that field only when the required Kummer excess rows are verified and the checked degree arithmetic remains representable. Finally, Ogdoad's partial **Ordinal** arithmetic need not pretend to construct Artin–Schreier roots that lie outside its verified window, even though the full- On_2 classification is mathematically complete.

1 Introduction and main result

Let F be a field of characteristic 2, let V be finite-dimensional, and let

$$Q : V \longrightarrow F, \quad Q(av) = a^2Q(v),$$

be a quadratic form. Its polar form is

$$B(u, v) = Q(u + v) + Q(u) + Q(v).$$

It is alternating. Write $R = \text{rad}(B)$, and call Q regular when $R = 0$. In repository terms this is an ordinary characteristic-2 (q, b) metric: the general-bilinear deformation coordinate `metric.a` is empty. The Artin–Schreier map and quotient are

$$\wp : F \longrightarrow F, \quad \wp(t) = t^2 + t, \quad F/\wp(F).$$

For a regular form and a symplectic basis $(e_i, f_i)_{i=1}^r$, the usual characteristic-2 Arf class is represented by

$$\sum_{i=1}^r Q(e_i)Q(f_i) \pmod{\wp(F)}.$$

Over a finite field this quotient has two elements and the absolute trace identifies it with $\mathbb{F}_2$. Over the full nimber field no finite absolute trace exists, so the intrinsic Artin–Schreier quotient is the correct starting point.

The answer is: *nothing replaces it over the full nimber field*. The class already lives naturally in $F/\wp(F)$, and this quotient is zero when $F = \text{On}_2$. More strongly, we give the complete singular normal form below.

The needed external input is standard. Conway's nim addition and nim multiplication turn the class of ordinals into the algebraically closed characteristic-2 field On_2 ; Lenstra states this explicitly and analyzes the algebraic-closure beginning segment [3]. All remaining arguments in this note are elementary linear algebra.

2 Elementary splitting lemmas

Lemma 2.1 (Closure consequences). *Let F be algebraically closed of characteristic 2. Then Frobenius $x \mapsto x^2$ and the Artin-Schreier map $\wp(x) = x^2 + x$ are both surjective. Hence F is perfect and $F/\wp(F) = 0$.*

Proof. For $a \in F$, algebraic closure supplies a root of $X^2 - a$ and a root of $X^2 + X - a$. These are respectively a square root of a and a preimage of a under $\wp$. Frobenius is injective in every field, so the square root is unique. $\square$

Let H denote the hyperbolic plane: in a basis e, f ,

$$Q(e) = Q(f) = 0, \quad B(e, f) = 1.$$

Lemma 2.2 (Every regular plane is hyperbolic). *Let F be algebraically closed of characteristic 2. Every regular quadratic plane over F is isometric to H .*

Proof. Choose a symplectic basis e, f , so $B(e, f) = 1$, and put $a = Q(e)$, $b = Q(f)$. Thus

$$Q(xe + yf) = ax^2 + xy + by^2.$$

If $a = 0$, then e is a nonzero isotropic vector. If $a \neq 0$, choose $t \in F$ satisfying

$$t^2 + t + ab = 0$$

and set $v = (t/a)e + f$. Direct substitution gives

$$Q(v) = \frac{t^2 + t + ab}{a} = 0.$$

In either case take a nonzero isotropic vector v . Regularity supplies w_0 with $B(v, w_0) = 1$. Since $Q(v) = 0$,

$$w = w_0 + Q(w_0)v$$

satisfies $B(v, w) = 1$ and

$$Q(w) = Q(w_0) + Q(w_0)B(w_0, v) = 0.$$

Thus v, w are a hyperbolic basis. $\square$

Proposition 2.3 (The regular part). *Every regular quadratic space of dimension $2r$ over an algebraically closed field of characteristic 2 is isometric to $H^{\perp r}$. Consequently $W_q(F) = 0$.*

Proof. An alternating nondegenerate form has even dimension and a symplectic plane. Lemma 2.2 makes the quadratic form on that plane hyperbolic. Its polar-orthogonal complement is again regular. Induction splits off r hyperbolic planes. Since every regular form is metabolic, its quadratic Witt class is zero. $\square$

The polar radical carries a small residue of information, but it is not an Arf class. Let Z denote the one-dimensional zero form and let A denote the one-dimensional form $x \mapsto x^2$; both have zero polar form.

Lemma 2.4 (The radical has only two forms). *Let R be an s -dimensional vector space over a perfect field of characteristic 2, and suppose the polar form of $Q|_R$ is zero. Then there is a unique linear functional $\ell : R \rightarrow F$ such that*

$$Q(r) = \ell(r)^2.$$

Up to isometry, $Q|_R$ is $Z^{\perp s}$ if $\ell = 0$, and $A \perp Z^{\perp(s-1)}$ if $\ell \neq 0$.

Proof. For a basis $r_1, \dots, r_s$, choose the unique $c_i \in F$ with $c_i^2 = Q(r_i)$ and define

$$\ell\left(\sum_i x_i r_i\right) = \sum_i c_i x_i.$$

Because the polar form vanishes,

$$Q\left(\sum_i x_i r_i\right) = \sum_i x_i^2 Q(r_i) = \left(\sum_i c_i x_i\right)^2.$$

Uniqueness follows from injectivity of Frobenius. A nonzero linear functional admits a basis in which it is the first coordinate; this gives the asserted normal forms. $\square$

3 Complete normal form over On_2

Theorem 3.1 (Transfinite number quadratic classification). *Let Q be a finite-dimensional quadratic form over the full number field On_2 , let B be its polar form, and write*

$$\text{rank}(B) = 2r, \quad \dim \text{rad}(B) = s, \quad \epsilon = \begin{cases} 0, & Q|_{\text{rad}(B)} = 0, \\ 1, & Q|_{\text{rad}(B)} \neq 0. \end{cases}$$

Then

$$(V, Q) \cong \begin{cases} H^{\perp r} \perp Z^{\perp s}, & \epsilon = 0, \\ H^{\perp r} \perp A \perp Z^{\perp(s-1)}, & \epsilon = 1. \end{cases}$$

Here the second case necessarily has $s \geq 1$. Thus two such forms are isometric over On_2 if and only if the triples $(2r, s, \epsilon)$ agree. In particular every regular form has a unique isometry class at fixed dimension, its Arf class is zero, and no transfinite Arf bit exists.

Proof. Choose a vector-space complement W to $R = \text{rad}(B)$. The restriction $B|_W$ is nondegenerate: if $w \in W$ is orthogonal to W , it is also orthogonal to R , hence lies in R , and therefore $w = 0$. Since $B(W, R) = 0$, the quadratic space is the orthogonal sum

$$(V, Q) = (W, Q|_W) \perp (R, Q|_R).$$

Proposition 2.3 identifies the first summand with $H^{\perp r}$, and Lemma 2.4 gives the two stated radical normal forms. Rank, radical dimension, and vanishing of $Q|_R$ are isometry invariants, so the normal forms are mutually distinct exactly when their triples differ. $\square$

Remark 3.2 (No hidden proper-class step). The theorem concerns a finite matrix of coefficients and performs finitely many field operations and root choices. The fact that On_2 is a proper class causes no extra algebraic step: every polynomial used in the proof has an ordinal root in Conway’s field. Explicitly, start with the set-sized subfield generated by the finitely many coefficients and take its relative algebraic closure inside On_2 ; this is a set-sized algebraically closed subfield in which the argument can be carried out.

The theorem says exactly what survives from the existing `ArfInvariants` report after scalar extension to full On_2 :

$$\text{arf} = 0, \quad \text{rank} = 2r, \quad \text{radical_dim} = s, \quad \text{radical_anisotropic} = \epsilon.$$

The last field means “ Q is nonzero on the polar radical”; it is not a remaining anisotropic regular core.

4 Why the finite Arf bit is still correct

There is no contradiction between Theorem 3.1 and the existing finite-nim-field classifier. Isometry depends on the ground field.

Let $K_d = \mathbb{F}_{2^d} \subset \text{On}_2$. For a regular form over K_d , its Arf class lies in

$$K_d/\wp(K_d) \cong \mathbb{F}_2,$$

where the isomorphism is the absolute trace. A form of nonzero Arf class is not hyperbolic over K_d , but it becomes hyperbolic after scalar extension to K_{2d} .

Proposition 4.1 (Every finite Arf bit dies one stage later). *The scalar-extension map*

$$W_q(K_d) \longrightarrow W_q(K_{2d})$$

is zero. Hence the directed colimit of the finite-field quadratic Witt groups inside On_2 is zero.

Proof. For $c \in K_d$, the polynomial $X^2 + X + c$ either already has a root in K_d or is an irreducible quadratic and therefore has a root in K_{2d} . Thus every c lies in $\wp(K_{2d})$, so the induced map $K_d/\wp(K_d) \rightarrow K_{2d}/\wp(K_{2d})$ is zero. Arf’s classification then makes the Witt-group map zero. Every element at every finite stage therefore maps to zero at a later stage, which is precisely the vanishing of the directed colimit. $\square$

For the smallest example, the plane $Q(e) = Q(f) = B(e, f) = 1$ is anisotropic over $\mathbb{F}_2$. If $t^2 + t + 1 = 0$ in $\mathbb{F}_4$, then $te + f$ is isotropic, so the same plane is hyperbolic over $\mathbb{F}_4$. The finite Arf bit records failure to split over the declared finite field; it was never invariant under arbitrary number scalar extension.

This settles the proposed “direct-limit finite-subfield invariant”: the direct limit is canonically zero. It also distinguishes two legitimate questions:

- Over the ideal full scalar world motivating `Ordinal`, namely On_2 , Theorem 3.1 is the complete classification.
- If one deliberately freezes the smallest subfield K generated by the coefficients and restricts all basis changes to $\text{GL}_n(K)$, the relative invariant belongs to $K/\wp(K)$. For transfinite coefficients this K need not be finite or algebraically closed. This Arf class is generally not by itself a complete higher-dimensional isometry invariant over such a field. That is a different, explicitly relative classification problem; it is not a scalar-level invariant of On_2 and has no canonical reduction to one bit.

Ogdoad already makes the first distinction for finite metrics: the raw `ArfInvariants` result is evaluated at a separately detected degree, and the higher `ClassifyWitt/ClassifyBrauerWall` surfaces store that degree alongside the Arf bit. The degree is part of those classes, not decoration.

5 The Clifford and Brauer–Wall boundary

The quadratic Witt conclusion immediately answers the Clifford half of the problem, with one scope qualification.

Corollary 5.1 (The regular Clifford class splits). *For every regular finite-dimensional quadratic form Q over On_2 , the graded Morita class of $\text{Cl}(Q)$ on Ogdoad’s characteristic-2 Clifford/Brauer–Wall surface is zero.*

Proof. By Theorem 3.1, $Q \cong H^{\perp r}$. Orthogonal sum maps to graded tensor product of Clifford algebras. For a hyperbolic basis e, f , the relations

$$e^2 = f^2 = 0, \quad ef + fe = 1$$

identify $\text{Cl}(H)$ with the graded matrix algebra on a $1|1$ -dimensional space, which is graded Morita-trivial. Therefore

$$[\text{Cl}(Q)] = r[\text{Cl}(H)] = 0.$$

□

This corollary concerns the subgroup of the graded Brauer–Wall theory carried by Clifford algebras of regular quadratic forms—the exact surface used by the repository’s finite characteristic-2 classifier. It does not claim a classification of every possible graded central simple algebra in characteristic 2. For singular Q , a nonzero polar-radical vector becomes a non-scalar homogeneous graded-central Clifford element (the characteristic-2 graded sign is trivial); if its square is zero, it also generates a nilpotent graded ideal. Thus the full Clifford algebra is not graded central simple on the required surface, so there is no Brauer–Wall class of the whole singular algebra; any polar-nondegenerate complement nevertheless has the split class above. Such a complement need not define a canonical quotient quadratic form when $Q|_{\text{rad}(B)} \neq 0$. The radical normal-form flag ϵ is quadratic-form data, not a Brauer–Wall coordinate.

6 Executable boundary

The mathematical ground field and the current Rust execution domain are not the same object.

- The intended scalar world is full On_2 , but `Ordinal::nim_mul` is implemented only through a source-verified Kummer window. Generic `Scalar::mul` is deliberately panic-on-escape.
- Mathematically, every ordinal below ω^{ω^ω} is algebraic over $\mathbb{F}_2$ and has finite degree. A finite metric there therefore lies in the compositum $\mathbb{F}_{2^m}$, where m is the least common multiple of the entry degrees. The repository’s `ordinal_common_finite_subfield_degree` certifies this field when every required Kummer excess is in the verified table and the checked degree arithmetic does not overflow, and returns `None` otherwise. Thus there is no mathematical third regime of entries in this segment that are not jointly contained in a finite nim-subfield, although the partial implementation can honestly report an operationally unknown degree.

- `arf_ordinal_finite` correctly classifies a metric over its detected and thereby declared common finite subfield, and uses that field’s absolute trace. It does not classify after scalar extension to the whole algebraic-closure segment. Its result is relative to the separately detected degree; the higher class wrappers record that degree, as Proposition 4.1 requires.
- A metric with coefficients outside the algebraic-closure initial segment, or an arbitrary metric over ideal full On_2 , is split over On_2 by Theorem 3.1; but constructing the isometry may require an Artin–Schreier root or square root outside the represented multiplication window. Returning a bare executable “ $\text{Arf} = 0$ ” without that distinction would confuse a mathematical base-extension theorem with a checked in-window witness.

Accordingly no new root-search oracle is required to close the research problem. A future implementation may expose a theorem-level `classification_over_full_on2` report, or a checked isometry only when all needed roots are representable. It should not invent a transfinite trace to $\mathbb{F}_2$.

7 Conclusion

The classification has four consequences:

1. **Domain.** The ideal full-field classification motivating the `Ordinal` scalar is over On_2 ; finite or coefficient-generated subfields are explicitly relative refinements.
2. **Invariant.** Over full On_2 , the Arf /Artin–Schreier quotient vanishes. The complete possibly singular invariant is $(\text{rank } B, \dim \text{rad } B, [Q]_{\text{rad } B} \neq 0)$.
3. **Root route.** Algebraic closure proves that the required roots exist. The proof gives explicit basis changes once such roots are supplied; executable root search remains bounded by the backend rather than by mathematics.
4. **Brauer–Wall.** Every regular quadratic Clifford class is split; singular radicals carry no class of the same kind.

The full-nimber theory therefore closes by collapse, not by a new transfinite bit. The finite bit remains useful precisely because it remembers a smaller declared ground field.

References

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